BGIN × agentprivacy

Collaborative Framework Proposal

Incident Response Agent Case Study

IEEE 7012 Data Contribution Integration

Foundation ↔ Labs ↔ DAO Sustainability Model

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Executive Summary

This proposal establishes a collaborative framework between the Blockchain Governance Initiative Network (BGIN) and agentprivacy.ai, using the Incident Response Agent project as a foundational case study. The framework demonstrates how privacy-preserving multi-agent architectures can enable secure information sharing between organizations while maintaining stakeholder sovereignty.

What is agentprivacy? agentprivacy (agentprivacy.ai) is a dual-agent protocol and architecture—a governance framework and living documentation system for agentic learning. It is not a wallet system. It exists in its own category, providing the agreement layer and governance scaffolding upon which wallet systems and other infrastructure can layer. This is why we begin with the agreement layer: establishing the governance foundation first enables interoperability with diverse wallet implementations, identity systems, and data infrastructure.

A Paradigm Shift in Knowledge Architecture: This project represents a fundamental shift in how knowledge graphs and system architecture are formed. In the same way that Claude learns skills through structured documentation, future more powerful AI systems will be able to ingest the agentprivacy living documentation—a continuously evolving story—and construct personal AI with foundational information theory already embedded. All the work defining private, self-sovereign AI to date becomes available to these systems: contributed and maintained by the ecosystem that forms around the duality of projection and protection of information, inclusive of all the technology that makes this possible, in a plethora of implementations.

The collaboration is structured around three pillars: (1) validating the agentprivacy governance framework through practical implementation, (2) establishing IEEE 7012-aligned data contribution and relationship agreements for agentic systems, and (3) creating a sustainable Foundation ↔ Labs ↔ DAO model that enables merit-based reciprocity while respecting BGIN's non-compensatory structure.

The technical foundation combines provenance graph infrastructure from NEC/Cybersmart with a knowledge graph built collaboratively by BGIN and its members. Both graphs remain iterative—continuously refined through the Block-to-Block development cycle. agentprivacy offers the origination of the knowledge graph as an initial in-kind contribution, with subsequent contributions from participating organizations aggregated between Block milestones and committed upon Block reviews.

Beyond the core framework, this collaboration creates pathways for joint pursuit of Ethereum Foundation grants, particularly in Privacy Pools and privacy authentication research. Success in these areas can lead to deeper collaboration and progressive decentralization—distributing both grant outcomes and governance authority to sustaining organizations and individual contributors who maintain agreements and permissions within the system.

This model positions both organizations to contribute to standards bodies, pursue collaborative grants, and potentially establish a maintained Labs entity for ongoing public goods stewardship—with contribution recognition mechanisms aligned to the Block-to-Block development lifecycle.

1. What is agentprivacy—and Why It's Different

1.1 A Dual-Agent Protocol and Architecture

agentprivacy (agentprivacy.ai) occupies a distinct category in the decentralized infrastructure landscape. It is a dual-agent protocol—architecting the relationship between personal agents (sovereign to individuals) and governance agents (operating with organizational permissions). It is not a wallet, not a credential issuer, and not a data storage system. Rather, it is:

A Governance Framework: Defining how agents, organizations, and individuals establish, maintain, and evolve their data relationships through formalized agreements.

Living Documentation: A continuously evolving knowledge base that captures governance patterns, regulatory mappings, and best practices for agentic systems—learning and adapting through each implementation.

An Agreement Layer: The foundational substrate upon which wallet systems, identity providers, and data infrastructure can build, ensuring governance coherence across diverse technical implementations.

A Dual-Agent Architecture: Formalizing the duality between projection (sharing, collaborating, contributing) and protection (privacy, sovereignty, consent) of information at the protocol level.

1.2 A Paradigm Shift in Knowledge Architecture

This project represents a fundamental shift in how knowledge graphs and system architecture are formed. Consider how Claude learns skills through structured documentation—SKILL.md files that encode best practices, workflows, and domain expertise. agentprivacy extends this pattern to an ecosystem scale:

AI-Native Documentation: The agentprivacy living documentation is structured as a continuously evolving story that future, more powerful AI systems can ingest. These systems will be able to construct personal AI with foundational information theory already embedded—privacy-preserving patterns, consent frameworks, and governance architectures baked in from initialization.

Ecosystem-Contributed Knowledge: All the work defining private, self-sovereign AI to date becomes available to these future systems: contributed and maintained by the ecosystem that forms around the duality of projection and protection of information.

Technology-Inclusive: The living documentation encompasses all the technology that makes privacy-preserving agentic systems possible—zero-knowledge proofs, verifiable credentials, decentralized storage, cryptographic key management—in a plethora of implementations that can be mixed, matched, and evolved.

1.3 Why Start with the Agreement Layer

By establishing governance foundations first, this approach enables:

Wallet Agnosticism: Any compliant wallet system can layer on top of the agentprivacy framework, inheriting its governance guarantees without vendor lock-in.

Identity Pluralism: Multiple identity systems (WorldID, government credentials, organizational attestations) can participate in the same governance framework.

Progressive Enhancement: Organizations can adopt the agreement layer immediately, then integrate technical infrastructure (wallets, storage, agents) as capabilities mature.

Regulatory Alignment: Governance-first design ensures compliance considerations are embedded from the start, rather than retrofitted onto technical systems.

2. Case Study: Incident Response Agent

2.1 Framework Validation

The BGIN Incident Response Agent provides an ideal proving ground for the agentprivacy governance framework. Information security incidents require rapid, coordinated response across organizational boundaries while maintaining strict confidentiality about vulnerabilities, affected parties, and remediation strategies.

The agentprivacy architecture addresses this through its dual-agent model: governance agents operate with organizational permissions while personal agents maintain individual sovereignty. For incident response, this translates to:

Component	Function in Incident Response
Archive Agent	Maintains encrypted historical incident data with selective disclosure capabilities for authorized parties
Codex Agent	Encodes response protocols, compliance requirements, and cross-jurisdictional coordination rules
Discourse Agent	Facilitates stakeholder communication while preserving appropriate confidentiality boundaries

2.2 Proof of Framework Efficacy

This case study will demonstrate several key capabilities:

Privacy-Preserving Information Sharing: Agents can share actionable intelligence about threats without revealing organizational vulnerabilities or affected user identities.

Verifiable Credential Exchange: Response team members authenticate through zero-knowledge proofs of organizational membership and role authorization.

Auditable Decision Trails: All agent actions create cryptographically verifiable logs for post-incident review and regulatory compliance.

Cross-Organizational Coordination: Multiple BGIN members can coordinate response efforts without exposing proprietary security postures.

3. IEEE 7012 MyTerms Agreement Framework

The agentprivacy architecture integrates IEEE 7012 agreements as a two-tier permission layer for all interactions with BGIN's agentic infrastructure. This creates a legally coherent framework for data dignity in agentic systems, distinguishing between ongoing service relationships and specific data contribution events.

3.1 Tier 1: Service Delivery Base Agreements

These agreements establish the foundational relationship between individuals and BGIN's organizational infrastructure. They would be present on all BGIN-hosted websites and govern access to BGIN agents.

Agreement	Purpose & Application
SD-BASE	The base level service delivery relationship agreement. This is the default relationship agreement for any individual using an agent to interact with BGIN infrastructure. It establishes the ongoing digital relationship between user and organization, governing how the service operates, what data is necessary for service delivery, and the mutual obligations of both parties. Required for: accessing BGIN websites, interacting with BGIN agents, participating in working group platforms.
SD-BASE-DP	A variant of SD-BASE with the added requirement that the individual wishes to include a data portability request in their terms. Under SD-BASE-DP, users maintain their own instance of all data generated within the relationship. This enables: personal archives of contributions, portable reputation credentials, sovereign data vaults for AI agent authorization. Recommended for: active contributors, working group participants, researchers requiring data continuity.

3.2 Tier 2: Data Contribution Agreements

These agreements govern specific, purposeful data contributions within an established SD-BASE relationship. They support one-off or targeted data sharing for particular purposes.

Agreement	Traditional Use	Agentic Application
PDC-AI	Data access for AI training and operation	Governs how personal agents may share user data with governance agents for collective AI improvement
PDC-GOOD	Data for Good projects	Enables incident data sharing for ecosystem-wide security improvements without competitive disclosure

PDC-INTENT	Intention Economy enablement	Allows users to express security preferences and tolerance levels through their agents

3.3 Agreement Architecture for BGIN Infrastructure

The two-tier structure creates a clear separation of concerns:

Touchpoint	Required Agreement	Optional Add-ons
BGIN Websites	SD-BASE	SD-BASE-DP, PDC-AI
Archive Agent Access	SD-BASE	SD-BASE-DP, PDC-GOOD
Codex Agent Access	SD-BASE	SD-BASE-DP, PDC-AI
Discourse Agent Access	SD-BASE	SD-BASE-DP, PDC-INTENT
Incident Response Participation	SD-BASE + PDC-GOOD	SD-BASE-DP

3.4 MyTerms Integration & Proof of Humanity

The agentprivacy framework proposes building agentic permissions and authentication around the MyTerms model, where users establish persistent, portable terms for data contribution that their agents enforce automatically. This creates:

User Sovereignty: Individuals control how their data contributes to collective goods, with agents as enforcement mechanisms rather than intermediaries requiring trust.

Proof of Humanity Integration: WorldID and similar verification systems provide the identity anchor for both SD-BASE relationships and PDC contribution agreements, ensuring human principals authorize agent actions.

Verifiable Data Relationships: All service relationships and data contributions create auditable trails linking back to human authorization through cryptographic verification.

4. Technical Foundation: The Three-Graph Architecture

The Incident Response Agent architecture relies on three complementary graph structures—each with distinct origins, governance, and purpose. Together, they create a novel value discovery mechanism. All three graphs are designed to be iterative, continuously refined through the Block-to-Block development cycle. This is why we start with governance foundations: the technical infrastructure will evolve, but the governance layer provides stability.

4.1 Provenance Graph (NEC/Cybersmart)

The provenance graph infrastructure is contributed by NEC/Cybersmart, providing the cryptographic audit trail for all agent actions, data transformations, and decision pathways. This establishes the verifiable lineage required for regulatory compliance and post-incident forensics. The provenance graph answers: who did what, when, with what authorization, and what were the downstream effects. As an iterative system, the provenance graph schema evolves to accommodate new action types, compliance requirements, and cross-jurisdictional audit needs.

4.2 Knowledge Graph (BGIN + Members)

The knowledge graph—containing governance documentation, regulatory mappings, incident taxonomies, and response protocols—is built collaboratively by BGIN and its members. This represents the collective intelligence of the network and remains in continuous evolution.

Initial Contribution: agentprivacy offers the origination of the knowledge graph as an in-kind contribution, providing the foundational schema, initial regulatory mappings, and integration architecture. Other organizations may similarly contribute foundational elements.

Ongoing Contributions: Between Block milestone junctions (e.g., Block 13 → Block 14 → Block 15), participating organizations contribute knowledge graph enhancements: new regulatory updates, incident patterns, protocol refinements, and cross-jurisdictional mappings.

Block Review Commits: Contributions are aggregated and committed upon Block reviews (e.g., Block 15 review), creating versioned snapshots of the collective knowledge. This establishes the lifecycle for contribution tracking and subsequent recognition.

4.3 Promise Graph (agentprivacy Dual-Agent Structure)

The third graph emerges from agentprivacy's dual-agent architecture: a promise graph that maps the commitments between personal agents and governance agents for each individual member. This structure draws directly from Promise Theory—where autonomous agents make voluntary commitments about their own behavior rather than imposing obligations on others.

Each member's personal agent maintains promises about data sharing, consent boundaries, and participation terms. Governance agents maintain reciprocal promises about data handling, access controls, and service delivery. These bilateral promises form edges in a graph that captures the trust topology of the network.

4.4 The Three-Graph Overlay: Value Discovery

When overlaid, the three graphs create a powerful value discovery mechanism:

Promise Graph captures intent and commitment between agents—what value *could* flow.

Provenance Graph records actual actions and transformations—what value *did* flow.

Knowledge Graph contextualizes meaning and relationships—*why* value matters.

The intersection reveals where promises were kept (trust reinforcement), where contributions exceeded commitments (recognition opportunities), and where knowledge gaps exist (collaboration invitations). This three-graph overlay becomes the substrate for contribution-based recognition, progressive decentralization, and emergent network value that no single participant could create alone.

5. Grant Pathways & Progressive Decentralization

Beyond the core framework development, this collaboration creates significant opportunities for joint grant pursuit and deeper organizational integration. Success in these pathways leads to progressive decentralization of both outcomes and governance authority.

5.1 Ethereum Foundation Grant Opportunities

The collaboration between BGIN and agentprivacy is well-positioned to pursue Ethereum Foundation grants in two complementary areas:

Privacy Pools Research: The agentprivacy governance framework directly complements Privacy Pools implementations by providing the agreement layer for association set membership and deposit/withdrawal authorization. Joint research can advance both the cryptographic mechanisms and the governance scaffolding required for compliant, privacy-preserving financial systems.

Privacy Authentication: Zero-knowledge proof systems for authentication require governance frameworks that define what credentials are accepted, who can issue attestations, and how disputes are resolved. The IEEE 7012 agreement integration provides this governance substrate for privacy-preserving authentication systems.

5.2 Progressive Decentralization Model

Successful grant outcomes and framework maturation enable progressive decentralization across multiple dimensions:

Dimension	Decentralization Pathway
Grant Distribution	Initial grants flow through founding organizations (BGIN, agentprivacy), with increasing allocation to sustaining member organizations based on contribution metrics tracked through verifiable credentials.
Governance Authority	Decision-making progressively shifts from founding organizations to the Contributor DAO, with voting weight determined by sustained participation and knowledge graph contributions.
Individual Contributors	Individual contributors who maintain agreements and permissions within the system accumulate portable credentials that grant access rights and recognition shares, independent of organizational affiliation.
Technical Infrastructure	Maintenance and hosting of knowledge graphs, provenance systems, and agent infrastructure progressively distribute across multiple sustaining organizations, reducing single points of failure.

5.3 Sustaining Organizations & Individual Access

The progressive decentralization model ensures that both organizations and individuals benefit from sustained participation. Organizations that contribute resources, expertise, or infrastructure become sustaining members with corresponding governance rights and grant allocation eligibility. Individual contributors who maintain agreements and permissions—whether through code contributions, documentation, testing, or governance participation—accumulate credentials that persist across organizational changes. This creates a resilient ecosystem where value accrues to contributors at all levels, not just founding entities.

6. Foundation ↔ Labs ↔ DAO Sustainability Model

Recognizing that BGIN, as a multi-stakeholder governance initiative, cannot directly compensate participants for project work, this proposal outlines a sustainability model that creates merit-based reciprocity through collaborative outcomes.

6.1 Collaborative Structure

Entity	Role	Value Flow
BGIN (Foundation)	Governance, standards, convening, legitimacy	Receives contribution-based recognition; provides membership network effects
agentprivacy Labs	Development, maintenance, commercialization	Revenue from enterprise implementations; funds continued open-source development
Contributor DAO	Merit tracking, contribution recognition, governance	Tracks contributions via verifiable credentials; enables participant ownership
Sustaining Orgs	Infrastructure, expertise, regional presence	Eligible for grant distribution; governance participation; network effects

6.2 Contribution Recognition Mechanics

When technologies developed through this collaboration generate commercial value, a contribution recognition and revenue sharing mechanism will apply. The participating roles include:

Open Source Foundation: Allocation for continued development of core public goods, managed transparently through the Contributor DAO.

Labs Operations: Allocation to sustain the commercial entity providing enterprise support, custom implementations, and ongoing maintenance.

Contributor Pool: Allocation distributed to contributors based on merit-weighted participation in development, documentation, testing, and governance.

Sustaining Organizations: Allocation to organizations that provide ongoing infrastructure, regional expertise, or specialized capabilities to the network.

> **Note:** Specific allocation percentages and distribution mechanics are proposed for discussion at the BGIN Steering Committee during Block 14. This allows stakeholder input on fair value distribution before formalizing the agreement.

6.3 Block-Aligned Contribution Lifecycle

Contributions follow the BGIN Block cadence, creating a natural lifecycle for recognition:

Phase	Activity
Block N	Contribution window opens; organizations submit knowledge graph updates, code, documentation
Inter-Block Period	Contributions aggregated; peer review; integration testing; credential issuance for verified contributions
Block N+1 Review	Contributions committed to versioned knowledge graph; contribution metrics finalized; recognition distributed

6.4 Contribution Tracking

Contributions are tracked through verifiable credentials issued for: code commits and reviews, documentation and specification writing, testing and security audits, governance participation in working groups, knowledge graph contributions (regulatory mappings, incident taxonomies, protocol updates), community engagement and standards body representation, and infrastructure provision by sustaining organizations. These credentials accumulate into reputation scores that determine recognition allocation, creating sustainable incentives for ongoing participation without requiring BGIN to directly compensate contributors.

7. Implementation Pathway

Phase 1: Case Study Development (Block 13 → Block 14)

Build and deploy the Incident Response Agent within the BGIN IKMG Working Group. Implement SD-BASE agreements on BGIN infrastructure as proof of concept. agentprivacy delivers initial knowledge graph as in-kind contribution. Demonstrate the framework's capability for sensitive information sharing between member organizations.

Phase 2: Standards Integration & Grant Preparation (Block 14 → Block 15)

Draft specifications for IEEE 7012 integration with agentic systems, covering both SD-BASE relationship agreements and PDC contribution agreements. Present findings to relevant standards bodies. Begin preparation of joint Ethereum Foundation grant applications for Privacy Pools and privacy authentication research. Aggregate inter-Block contributions for Block 15 review and commit.

Phase 3: Grant Applications & Ecosystem Expansion (Q3-Q4 2026)

Submit collaborative grant proposals to organizations including: Ethereum Foundation (Privacy Pools, privacy authentication), Protocol Labs (decentralized infrastructure), World Foundation (proof of humanity integration), and relevant government research programs (US NSF, Japan JST). Begin onboarding sustaining organizations.

Phase 4: Labs Formation & Progressive Decentralization (2027)

Establish formal Labs entity if commercial interest warrants, with governance structure and contribution recognition mechanisms as outlined in Section 6. Begin progressive decentralization of grant distribution and governance authority to sustaining organizations and individual contributors based on accumulated credentials and participation metrics.

8. Proposal Terms

This proposal requests:

From BGIN: Working group participation venue, membership network access for case study collaboration, infrastructure for SD-BASE agreement deployment, standards body coordination, and recognition of contributions through existing governance mechanisms.

From agentprivacy: Technical development of the Incident Response Agent, IEEE 7012 integration specifications (both SD-BASE and PDC variants), initial knowledge graph origination as in-kind contribution, framework documentation, and commitment to open-source delivery of all collaborative outputs.

From NEC/Cybersmart: Provenance graph infrastructure contribution for verifiable audit trails and compliance documentation.

Mutual Commitment: Joint pursuit of collaborative grants (particularly Ethereum Foundation Privacy Pools and privacy authentication), shared authorship on standards contributions,

Block-aligned contribution tracking, and good-faith negotiation of Labs formation and progressive decentralization terms—with revenue sharing mechanics to be discussed at Block 14 Steering Committee.

Appendix A: Steering Committee Brief

Proposed Framework Updates for AI Agent Collaboration

> As BGIN develops AI-powered tools for governance documentation and incident response, we propose introducing IEEE 7012 agreements as our permission framework. Importantly, agentprivacy is not a wallet system—it is a governance framework and living documentation for agentic learning, existing in its own category. This is why we start with the agreement layer (SD-BASE/SD-BASE-DP for ongoing relationships, PDC-AI/PDC-GOOD/PDC-INTENT for specific contributions), then allow wallet systems to layer on top. The technical foundation combines provenance graph infrastructure from NEC/Cybersmart with a knowledge graph built collaboratively by BGIN members—both designed to be iterative. agentprivacy offers initial knowledge graph origination as an in-kind contribution. This Block-aligned contribution lifecycle (contribute → aggregate → commit at Block review) enables continuous improvement while maintaining transparent contribution tracking. The collaboration also creates pathways for joint Ethereum Foundation grants in Privacy Pools and privacy authentication, leading to progressive decentralization of both grant distribution and governance authority to sustaining organizations and individual contributors. Specific revenue sharing mechanics are proposed for discussion at Block 14 Steering Committee.

Appendix B: Sponsor Engagement Language

For Sustaining Sponsor Communications (e.g., World Foundation)

> BGIN's governance documentation and incident response initiatives are building agentic systems with proof-of-humanity verification and IEEE 7012 data agreements at their foundation—ensuring human accountability remains central to AI agent networks. Sponsoring this work provides direct reciprocity: your verification infrastructure becomes the identity layer for privacy-preserving governance agents, while collaborative standards development creates interoperable frameworks that expand ecosystem adoption. Our framework provides the governance substrate upon which wallet systems layer, creating the legal and technical scaffolding for sustainable data dignity at scale. Joint pursuit of Ethereum Foundation grants in Privacy Pools and privacy authentication creates additional value—successful outcomes progressively decentralize to sustaining organizations and individual contributors who maintain agreements within the system. This is public goods development where contribution generates compounding returns through network effects.

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